IOT BASED DIGITAL LCD NOTICE BOARD

An
Industrial Oriented Mini Project

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology
in
Computer Science and Engineering

Submitted by

GANDAMALLA ANJALI	23SS1A0518
GHANAPURAM DHARANI	23SS1A0520
KASOJU POOJITHA	23SS1A0531
PAMMI KOMALI KAVYA	23SS1A0548

Under the guidance of

Mrs. K. Neeraja

Assistant Professor

Department of Computer Science and Engineering

JNTUH University College of Engineering Sultanpur

Sultanpur(V), Pulkal(M), Sangareddy district, Telangana-502273

April, 2026

JNTUH UNIVERSITY COLLEGE OF ENGINEERING SULTANPUR

Sultanpur(V), Pulkal(M), Sangareddy-502273, Telangana

Department of Computer Science and Engineering

Certificate

This is to certify that an Industrial Oriented Mini Project report work entitled"**IOT BASED DIGITAL LCD NOTICE BOARD** "is a bonafide work carried out by a team consisting of **GANDAMALLA ANJALI** bearing Roll no. **23SS1A0518**, **GHANAPURAM DHARANI** bearing Roll no. **23SS1A0520**, **KASOJU POOJITHA** bearing Roll no. **23SS1A0531**, **PAMMI KOMALI KAVYA** bearing Roll no. **23SS1A0548**, in partial fulfillment of the requirements for the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING discipline to Jawaharlal Nehru Technological University Hyderabad Unviersity College of Engineering Sultanpur during the academic year 2025 - 2026.

The results embodied in this report have not been submitted to any other University or Institution for the award of any degree or diploma.

Guide
Mrs . K . NEERAJA
Assistant Professor

HOD
Dr . P . SWETHA
Professor

External Examiner

Declaration

We hereby declare that the Industrial Oriented Mini Project entitled "**IOT BASED DIGITAL LCD NOTICE BOARD**" is a bonafide work carried out by a team consisting of **GANDAMALLA ANJALI** bearing Roll no.**23SS1A0518**, **GHANAPURAM DHARANI** bearing Roll no .**23SS1A0520**, **KASOJU POOJITHA** bearing Roll no.**23SS1A0531**, **PAMMI KOMALI KAVYA** bearing Roll no. **23SS10548**, in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering discipline to Jawaharlal Nehru Technological University Hyderabad College of Engineering Sultanpur during the academic year 2025- 2026. The results embodied in this report have not been submitted to any other University or Institution for the award of any degree or diploma.

GANDAMALLA ANJALI **23SS1A0518**

GHANAPURAM DHARANI **23SS1A0520**

KASOJU POOJITHA **23SS1A0531**

PAMMI KOMALI KAVYA **23SS1A0548**

Acknowledgment

We wish to take this opportunity to express our deep gratitude to all those who helped us in various ways during our Industrial Oriented Mini Project report work. It is our pleasure to acknowledge the help of all those individuals who were responsible for foreseeing the successful completion of our Industrial Oriented Mini Project report.

We express our sincere gratitude to **Dr. S. Viswanadha Raju, Senior Professor and Principal**, JNTUHUCES for his support during the course period.

We sincerely thank **Dr. Y. Raghavender Rao, Professor and Vice Principal**, JNTUHUCES for his kind help and co-operation.

We sincerely thank **Dr. P. Swetha, Professor and HOD**, Computer Science and Engineering for her kind help and co-operation.

We are thankful to our Industrial Oriented Mini Project Coordinator **Ms. Shumama Ansa, Assistant Professor** and guide **Mrs. K. Neeraja, Assistant Professor** for their constant support, effective suggestions and guidance in the completion of our project.

Finally, we express our gratitude with great admiration and respect to our faculty for their moral support and encouragement throughout the course.

GANDAMALLA ANJALI	**23SS1A0518**
GHANAPURAM DHARANI	**23SS1A0520**
KASOJU POOJITHA	**23SS1A0531**
PAMMI KOMALI KAVYA	**23SS1A0548**

Contents

Abstract

In the evolving digital landscape of educational institutions, efficient and timely communication is essential for maintaining an organized and responsive environment. The IoT Based Digital LCD Notice Board emerges as a modern solution designed to replace traditional paper-based notice boards with a smart, automated, and internet-enabled system. Conventional notice boards often involve manual effort, delayed updates, and excessive paper usage, making them inefficient for real-time communication.The system integrates a web-based interface, a cloud database, and a Wi-Fi-enabled microcontroller such as ESP8266 or ESP32 to create a seamless communication platform. Authorized faculty members can securely log in and upload, update, or delete notices through an intuitive dashboard. These notices are stored in a centralized cloud database and retrieved wirelessly by the microcontroller, which displays them on a 16×2 LCD module using I2C communication. An automatic scrolling mechanism ensures that lengthy messages are displayed clearly.By eliminating manual posting and enabling instant updates, the system reduces administrative workload and improves operational efficiency. The centralized architecture allows multiple display units to be managed effectively, ensuring synchronized updates. Overall, the IoT Based Digital LCD Notice Board provides a reliable, scalable, and cost-effective solution for modern institutional communication.

List of Figures

List of Tables

Chapter 1

INTRODUCTION

Effective communication is a fundamental requirement in educational institutions and organizations. Traditional notice boards are widely used to share information; however, they depend on manual paper posting, physical presence, and repeated updates. This process is often time-consuming, inefficient, and not suitable for urgent announcements. Paper-based systems also lead to wastage of resources and lack centralized control.

With the advancement of the Internet of Things (IoT), physical devices can now be connected to the internet for real-time data exchange and automation. By integrating IoT technology into institutional communication systems, notices can be updated instantly and displayed digitally without manual effort.The IoT Based Digital LCD Notice Board is developed as a modern solution to overcome the limitations of traditional notice boards. It utilizes wireless communication and cloud storage to enable real-time information display, ensuring faster and more efficient communication.

1.1 Project Overview

The IoT Based Digital LCD Notice Board is a cloud-connected digital system that allows authorized users to manage and display notices electronically. The system consists of a web-based interface, a cloud database, a Wi-Fi-enabled microcontroller (ESP8266/ESP32), and a 16×2 LCD display.Faculty members log in to a secure dashboard to add, update, or delete notices. The notices are stored in a centralized cloud database such as Firebase. The ESP microcontroller connects to Wi-Fi and periodically retrieves the latest notice from the cloud using HTTP/REST communication. The fetched message is then displayed on the LCD screen using I2C communication. If the message exceeds the display size, an automatic scrolling feature ensures complete visibility.The system is designed to reduce manual effort, enable centralized control, improve communication speed, and promote a paperless environment.

Effective communication is essential in educational institutions, yet current notice management practices fail to meet modern demands. Traditional and partially digital systems lack real-time accessibility, centralized monitoring, and automated updating mechanisms. These limitations lead to delayed information sharing, inefficient administrative processes, and reduced communication reliability. There is a clear need for a secure, scalable, and automated notice display system that enables remote management, ensures instant updates, and supports efficient institutional communication.

1.2 Purpose

The purpose of the IOT Based Digital LCD Notice Board is to develop a smart and efficient communication system that replaces traditional paper-based notice boards with a digital solution. The system enables authorized users to create and update notices remotely through a secure web interface. Notices are stored in a centralized cloud database and displayed instantly on an LCD screen using a Wi-Fi enabled micro-controller. By automating the display process, the system reduces manual effort, minimizes delays, and ensures real-time information sharing. It also promotes a paperless environment, lowers operational costs, and provides centralized control for managing multiple display units efficiently.

1.3 Existing System

In many educational institutions, notice boards are managed using traditional paper-based methods. Announcements are printed and manually displayed, which increases workload and often leads to delays. Paper notices can be removed, damaged, or become outdated without timely updates. Some institutions use partially digital systems such as computer-connected or GSM-based boards, but these require local connections, depend on network availability, and may involve additional costs and limitations. Overall, existing systems lack centralized control, real-time updates, and scalability. These limitations create the need for an automated, internet-based notice system that ensures faster and more reliable communication.

Disadvantages of Existing System:

- **Manual Inefficiencies:** Traditional paper-based notice boards require printing, physical posting, and regular manual updates. This process consumes time, increases workload, and may lead to delays in communication.
- **Delayed Information Dissemination:** Important or urgent announcements may not reach students or staff immediately, resulting in miscommunication and inconvenience.
- **Paper Waste and Environmental Impact:** Frequent printing of notices leads to excessive paper usage, contributing to unnecessary resource consumption and environmental concerns.
- **Lack of Centralized Control:** Existing systems do not provide a unified platform to manage multiple notice boards, making monitoring and updating difficult.
- **Limited Remote Accessibility:** Many digital boards require local system connections or manual intervention, restricting remote management capabilities.

1.4 Proposed System

The proposed system is a cloud-integrated IoT communication platform designed to deliver announcements directly to a digital display unit without manual intervention. The system architecture consists of three major components: a web-based control panel, a centralized cloud database, and a Wi-Fi-enabled microcontroller connected to an LCD module.Administrators access a secure dashboard where notices can be managed dynamically. Each notice is stored with relevant metadata such as timestamp and priority. The ESP8266/ESP32 module continuously monitors the cloud database for updates. When a new notice is detected, the device retrieves the data and updates the display automatically without requiring system restart or physical access.The LCD module is interfaced using I2C communication to reduce wiring complexity and ensure stable data transmission. The firmware running on the microcontroller handles message parsing, formatting, and scrolling logic to optimize visibility within the limited display size.The system is designed with scalability in mind, allowing multiple display units to synchronize with the same database. It ensures secure communication through authenticated access and structured data handling. The modular design also enables future integration with mobile notifications, larger display panels, or multimedia extensions.This proposed solution emphasizes automation, centralized monitoring, low hardware complexity, and reliable wireless communication to enhance institutional announcement management.

1.5 Conclusion

The analysis of traditional notice display methods highlights several operational limitations, including manual effort, delayed updates, and lack of centralized management. These challenges create inefficiencies in institutional communication and demonstrate the necessity for a more structured and technology-driven approach.The IoT Based Digital LCD Notice Board presents a practical solution by integrating cloud connectivity, wireless communication, and embedded systems into a unified platform. By enabling secure remote notice management and real-time display through a Wi-Fi-enabled microcontroller, the system enhances reliability, speed, and accessibility of information dissemination.The proposed solution establishes a strong foundation for developing a scalable and automated communication system that aligns with modern digital infrastructure. The subsequent sections further detail the system architecture, design methodology, and implementation aspects to demonstrate the feasibility and effectiveness of the project.

Chapter 2

LITERATURE SURVEY

2.1 Introduction

The rapid growth of Internet of Things (IoT) technology has significantly transformed communication systems in educational institutions and organizations. Traditional notice boards rely on manual updates, which are time-consuming and inefficient. Recent studies focus on integrating embedded systems, wireless communication, and cloud platforms to develop smart and automated notice display systems. This literature survey reviews existing research related to IoT-based digital notice boards, cloud-integrated communication systems, and Wi-Fi-enabled microcontroller applications.

- **1. IoT-Based Smart Notice Board Systems:**
  Several research works highlight the development of IoT-enabled notice boards using microcontrollers like ESP8266 and ESP32. These systems allow real-time updates of notices through internet connectivity, reducing manual effort and improving communication efficiency.

- **2. Wi-Fi Enabled Microcontrollers in Embedded Systems:**
  Studies on ESP8266 and ESP32 demonstrate their effectiveness in wireless data transmission. Their built-in Wi-Fi capability enables seamless communication between cloud databases and display modules, making them suitable for smart campus applications.

- **3. Cloud Database Integration:**
  Research emphasizes the use of cloud platforms such as Firebase for real-time data syn-

chronization. Cloud databases allow centralized storage and remote access, enabling multiple display units to retrieve updated information instantly.

- **4. LCD and I2C Communication Modules:**
  Existing implementations utilize 16×2 LCD displays with I2C modules to reduce wiring complexity and enhance communication stability. I2C protocol ensures efficient data transfer between the microcontroller and display unit.

- **5. Web-Based Notice Management Systems:**
  Many projects integrate web interfaces developed using HTML, CSS, and JavaScript to allow authorized users to post and manage notices securely. This enhances usability and remote accessibility.

- **6. Secure and Scalable IoT Communication:**
  Recent studies focus on improving system reliability, secure authentication, and scalability in IoT communication platforms. These improvements ensure safe data transmission and support multiple connected devices.

2.2 Conclusion

The reviewed literature demonstrates that IoT-based digital notice board systems effectively address the limitations of traditional paper-based communication methods. By combining Wi-Fi-enabled microcontrollers, cloud databases, LCD display modules, and web interfaces, modern systems provide real-time updates, centralized control, and improved communication efficiency. The IoT Based Digital LCD Notice Board project builds upon these advancements to develop a reliable, scalable, and automated institutional communication solution.

Chapter 3

Requirement Specification

3.1 Software Requirement

- **Functional Requirements :**
  Graphical User Interface with the User.
- **FrontEnd Development :**
  HTML, CSS3 (for styling), JavaScript (for interactive features).
- **Backend Development :**
  Python (Flask) / Node.js for web server development.
- **Cloud Database :**
  Firebase (Real-Time Database).
- **IoT Programming :**
  Embedded C / Arduino Programming Language (for ESP8266 / ESP32).
- **Authentication :**
  Username Login with Password.
- **Operating System :**
  Windows 7 (32-bit) or above.
- **IDE :**
  VS Code, Arduino IDE.

3.2 Hardware Requirement:

- **Processor :**
  Minimum Intel Core i3 (for development and programming).
- **RAM :**
  Minimum 4 GB.
- **Hard Disk :**
  Minimum 20 GB free space (for IDE, libraries, and project files).
- **Development System :**
  Computer / Laptop for programming and firmware upload.
- **Microcontroller :**
  ESP8266 (NodeMCU) / ESP32 with built-in Wi-Fi.
- **Display Module :**
  16×2 LCD Display with I2C Module.
- **Internet Connection :**
  Wi-Fi Internet connection for cloud communication.
- **Connecting Components :**
  Jumper wires for hardware connections.
- **Power Supply :**
  5V USB Cable / Power Adapter for ESP module.
- **Cloud Server :**
  Firebase (for real-time data hosting).

3.3 Conclusion

In conclusion, the defined hardware and software requirements provide a stable foundation for the IoT Based Digital LCD Notice Board. The development system ensures smooth programming and testing, while the ESP8266/ESP32 microcontroller with the 16×2 LCD module enables real-time notice display through Wi-Fi connectivity. The use of Arduino IDE for embedded programming and Firebase for cloud database management supports efficient communication and seamless data synchronization. Together, these components ensure reliable performance, centralized control, and continuous operation of the system.

Chapter 4

WORKING OF SYSTEM

4.1 SYSTEM ARCHITECTURE

The system architecture provides an overall view of the IoT Based Digital LCD Notice Board and explains how notices are created, stored, and displayed in real time. The working of the system is described as follows:

1. Faculty/Staff Web Page (Add / Update / Delete):
Authorized faculty or staff members access a web page to manage notices. Using this interface, they can add new notices, update existing notices, or delete outdated notices.

2. Internet (Wi-Fi):
The web page sends the notice data through the internet using Wi-Fi. This enables remote updating without any physical interaction with the notice board.

3. Cloud Database (Firebase):
All notices are stored in a centralized cloud database (Firebase). This acts as the central notice repository and ensures that the latest notice is available at any time for the display device.

4. Wi-Fi Communication (HTTPS/REST):
The ESP8266/ESP32 connects to the cloud database using Wi-Fi and communicates through HTTPS/REST APIs (or Firebase APIs) to fetch the latest notice securely.

5. ESP8266 / ESP32 Wi-Fi Client Controller:

The ESP8266/ESP32 continuously checks the cloud database for updated notices. When a new notice is detected, it retrieves the latest notice data and prepares it for display.

6. 16×2 LCD Display (I2C) + Scrolling:

The fetched notice is displayed on a 16×2 LCD module interfaced through I2C communication. If the notice length exceeds the screen width, a scrolling mechanism is applied to ensure the complete message is visible.

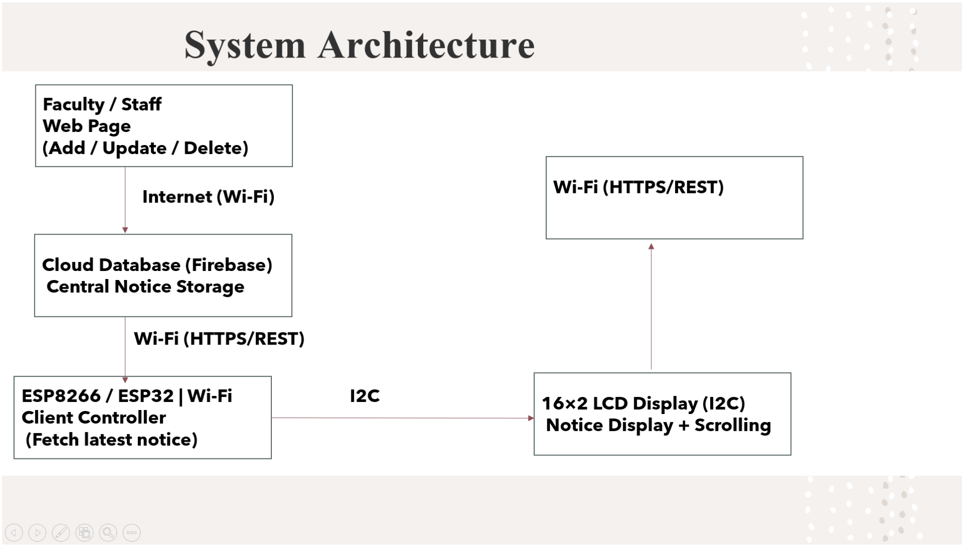

Figure 4.1: *System Architecture of IoT Based Digital LCD Notice Board*

4.1.1 Working Steps

- Faculty/Staff logs in to the web page and enters the notice.
- Notice is uploaded to Firebase cloud database through Wi-Fi.
- ESP8266/ESP32 connects to Wi-Fi and fetches the latest notice using HTTPS/REST.
- The notice is displayed on the 16×2 LCD using I2C.
- Long notices scroll automatically for full visibility.

4.1.2 Conclusion

The IoT Based Digital LCD Notice Board integrates a web interface, Firebase cloud storage, and a Wi-Fi-enabled microcontroller to provide real-time notice updates. This architecture reduces manual effort, avoids paper usage, enables centralized management, and ensures faster and more reliable communication within the institution.

Chapter 5

SYSTEM DESIGN

Design is the abstraction of a solution. It is a general description of the solution to a problem without the detailed implementation. Design helps transform patterns identified during the analysis phase into a structured model during the design phase. After the design phase, the time required to create the implementation can be reduced.

A UML diagram is a diagram based on the UML (Unified Modeling Language) with the purpose of visually representing a system along with its main actors, roles, actions, artifacts, or classes, in order to better understand, alter, maintain, or document information about the system.

5.1 UML DIAGRAMS

What is UML?
UML is an acronym that stands for Unified Modelling Language. Simply put, UML is a modern approach to modelling and documenting software. In fact, it is one of the most popular business process modelling techniques.

It is based on diagrammatic representations of software components. As the old proverb says: “a picture is worth a thousand words”. By using visual representations, we are able to better understand possible flaws or errors in software or business processes.

Building Blocks of UML: The vocabulary of UML encompasses three kinds of building blocks.

- **Things:** Things are the abstractions that are first-class citizens in a model.
- **Relationships:** Relationships tie these things together.
- **Diagrams:** Diagrams group interesting collections of things.

5.2 Class Diagram

This class diagram illustrates the structural organization of the IoT Based Digital LCD Notice Board system by defining the relationships between its core entities. The User class represents authorized personnel responsible for initiating system actions. The Notice class encapsulates attributes related to announcement content and maintains the state of each message. The Controller (ESP Module) class functions as the intermediary that processes data and controls hardware operations. The Display Module class represents the LCD unit responsible for rendering formatted text output.

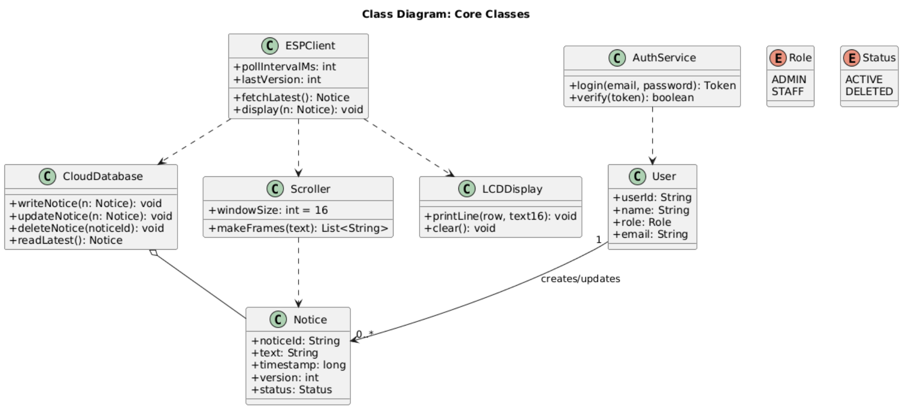

Figure 5.1: Class Diagram

5.3 Usecase Diagram

This diagram illustrates the interactions between users and the system. The Faculty/Admin can log in, create, update, and delete notices. The IoT device retrieves and displays the latest notice, while the system manages data storage and synchronization. It defines the system's functionalities from the perspective of external actors.

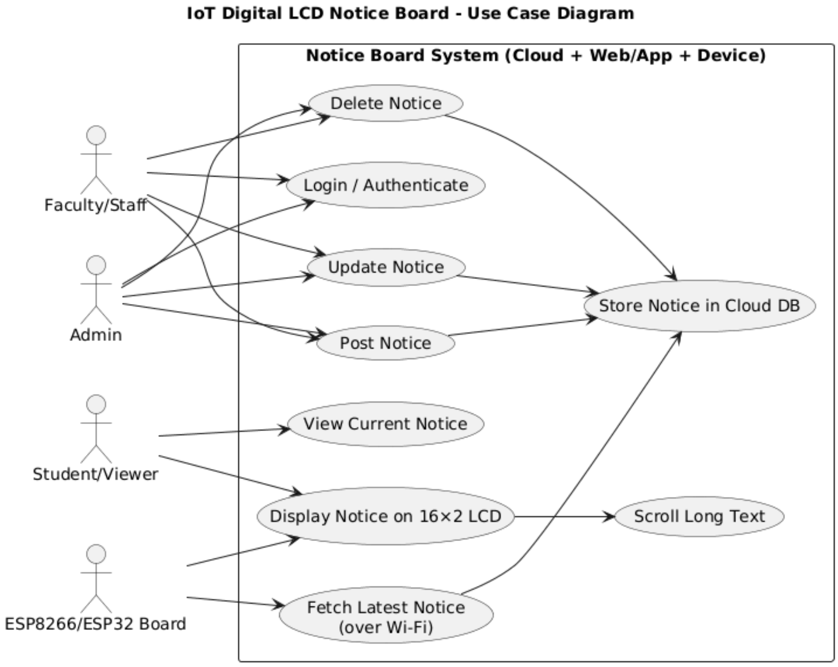

Figure 5.2: *Usecase Diagram*

5.4 Activity Diagram

This diagram outlines the workflow of posting and displaying a notice. It begins with user authentication, followed by notice creation or update, data validation, and storage in the cloud database. The IoT device then retrieves the updated notice, processes the message, and displays it on the LCD screen.

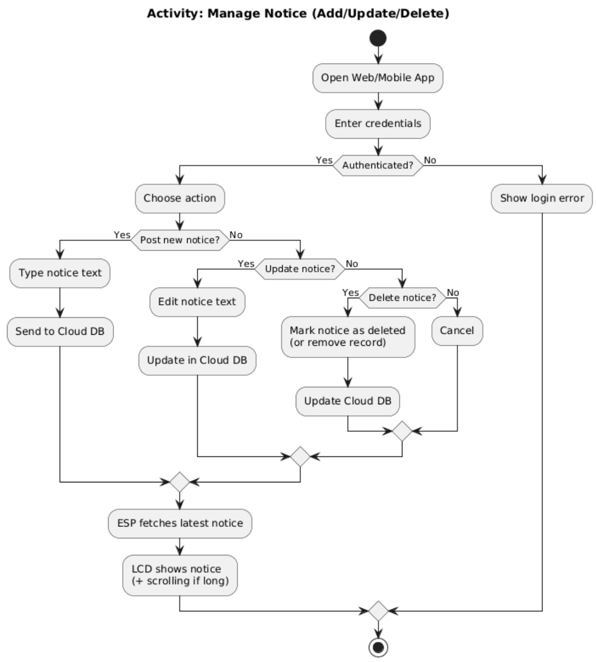

Figure 5.3: *Activity Diagram*

5.5 Sequence Diagram

The sequence diagram illustrates how different components of the system interact over time during the notice publishing process. It shows the sequence of messages exchanged between the Faculty/Admin, the Web Application, the Cloud Database, and the IoT Device (ESP8266/ESP32).

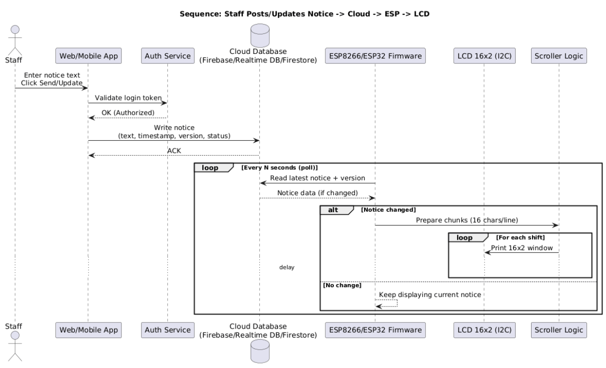

Figure 5.4: *Sequence Diagram*

5.6 Object Diagram

The object diagram represents a snapshot of the system at a particular moment, showing real instances of classes and the links between them. It illustrates how specific objects such as an Admin user, a Notice message, the Cloud Database entry, and the LCD Display unit are connected and hold values during runtime.

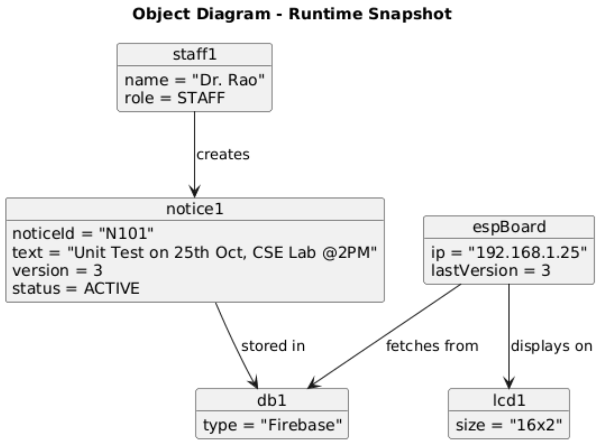

Figure 5.5: *Object Diagram*

5.7 Component Diagram

The component diagram represents the high-level structure of the system by showing how different modules are organized and interconnected. It illustrates major components such as the Web Application Interface, Authentication Module, Cloud Database, and the IoT Device (ESP8266/ESP32 with LCD Module).

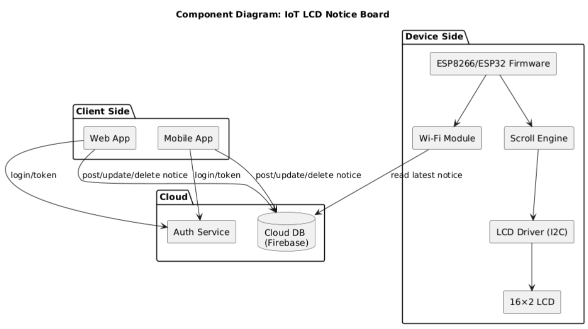

Figure 5.6: *Component Diagram*

5.8 Deployment Diagram

The deployment diagram illustrates the physical arrangement of hardware and software components in the system. It shows nodes such as the User Device (Browser), Web Server, Cloud Database (Firebase), and the IoT Hardware (ESP8266/ESP32 with LCD Module).

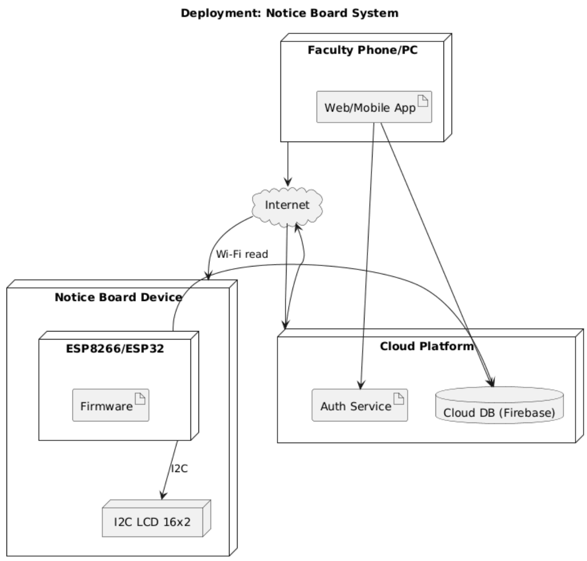

Figure 5.7: *Deployment Diagram*

5.9 Conclusion

The Unified Modeling Language (UML) provides a structured methodology for specifying and documenting software system designs. It includes multiple types of diagrams, each representing a different aspect of the system architecture and functionality. These diagrams collectively help developers and designers understand the system structure, interactions, and deployment environment.

Chapter 6

IMPLEMENTATION

6.1 Overview

The implementation phase focuses on converting the designed architecture into a functional system. The IoT Based Digital LCD Notice Board is implemented using a combination of web technologies, cloud services, and embedded hardware components. The system consists of a web interface for notice management, a cloud database for storing notices, and a Wi-Fi enabled microcontroller that retrieves the notices and displays them on an LCD module.

The frontend interface is developed using HTML and CSS, while Firebase is used as the cloud database for storing notices in real time. The ESP8266 microcontroller is programmed using Embedded C in the Arduino IDE to retrieve notice data through HTTP requests and display it on the LCD screen using I2C communication.

The implementation is divided into multiple modules to simplify development and improve maintainability.

6.2 Modules

The implementation of the system is divided into the following modules:

- Home Page Module
- LCD Integration Module
- Deployment Module

6.3 Home Page Module

The Home Page module acts as the entry point of the system. It provides a simple and clear interface through which users can understand the purpose of the Digital Notice Board and navigate to the next stage of operation. This page is designed to offer a clean layout with the project title and a direct login option for faculty members.

The home page is developed using HTML for structuring the content and CSS for styling and layout design. It ensures a responsive and visually organized interface so that users can easily interact with the system. The page presents the main heading of the application and a login button that redirects authorized faculty members to the authentication page for managing notices.

In addition to navigation, the home page introduces the overall concept of the system and helps users quickly understand the functionality of the digital notice board. A well-structured home page improves accessibility and usability by reducing confusion and providing clear directions for system interaction. Since it is the first screen viewed by users, this module plays an important role in creating a user-friendly impression and ensuring smooth access to the notice management system.

6.3.1 Home Page Code

```html
<!DOCTYPE html>
<html>
<head>
    <title>Digital Notice Board</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
    <h1>Digital Notice Board</h1>
</header>

<div class="main-container">
    <h2>Welcome to Digital Notice Board</h2>
    <a href="login.html" class="login-btn">Faculty Login</a>
</div>

</body>
</html>
```

The above code creates the basic structure of the home page. It contains a heading and a login button that directs faculty members to the login page for managing notices.

6.4 LCD Integration Module (Arduino IDE)

This module is responsible for fetching notices from Firebase and displaying them on the 16×2 LCD using the ESP8266 microcontroller. The microcontroller connects to Wi-Fi, retrieves the latest notices, and shows them on the LCD. If the notice is too long, scrolling is applied so the full text can be viewed.

6.4.1 ESP8266 Program Code

```
#include <ESP8266WiFi.h>
#include <ESP8266HTTPClient.h>
#include <ArduinoJson.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <WiFiClientSecureBearSSL.h>

// WiFi credentials
const char* ssid = "OnePlus Nord CE3 5G";
const char* password = "franzz33";

// Firebase REST URL
const String firebaseURL =
"https://firestore.googleapis.com/v1/projects/digital-notice-
   board-7e57f/databases/(default)/documents/notices";

// LCD configuration
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Secure client
BearSSL::WiFiClientSecure client;

// Scrolling variables
String scrollingText = "";
int scrollIndex = 0;

void fetchNotices();

void setup() {

  Serial.begin(115200);
  Wire.begin(D2, D1);

  lcd.init();
  lcd.backlight();
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("Connecting WiFi");

```

```
  WiFi.begin(ssid, password);

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
  }

  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("WiFi Connected");

  client.setInsecure();
  fetchNotices();
}

void loop() {

  if (scrollingText.length() > 16) {

    String displayText =
    scrollingText.substring(scrollIndex, scrollIndex + 16);

    lcd.setCursor(0,0);
    lcd.print(displayText);

    scrollIndex++;

    if (scrollIndex + 16 > scrollingText.length()) {
      scrollIndex = 0;
    }

  } else {

    lcd.setCursor(0,0);
    lcd.print(scrollingText);

  }

  static unsigned long lastFetch = 0;

  if (millis() - lastFetch > 30000) {
    fetchNotices();
```

```
    lastFetch = millis();
  }

  delay(300);
}

void fetchNotices() {

  if (WiFi.status() == WL_CONNECTED) {

    HTTPClient http;
    http.begin(client, firebaseURL);

    int httpCode = http.GET();

    if (httpCode > 0) {

      String payload = http.getString();

      const size_t capacity = 16 * 1024;
      DynamicJsonDocument doc(capacity);

      DeserializationError error =
      deserializeJson(doc, payload);

      if (!error) {

        scrollingText = "";

        JsonArray documents =
        doc["documents"].as<JsonArray>();

        int count = 0;

        for (JsonObject notice : documents) {

          if (count >= 4) break;

          String text =
          notice["fields"]["text"]["stringValue"];

```

```
          if (scrollingText.length() > 0)
            scrollingText += " | ";

          scrollingText += text;

          count++;
        }

        scrollIndex = 0;
        Serial.println(scrollingText);
      }
    }

    http.end();
  }
}
```

The above program enables the ESP8266 to fetch cloud-based notices and display them on the LCD screen in real time.

6.5 Deployment Module

The Deployment Module defines how the frontend files of the Digital Notice Board system are hosted and accessed through the internet. In this project, Firebase Hosting is used to publish the web application and make it available to users online. Firebase provides a reliable and secure platform for hosting static web files such as HTML, CSS, and JavaScript. It also ensures fast content delivery and secure communication using HTTPS.

By deploying the frontend through Firebase Hosting, the web interface can be accessed from any device with an internet connection. This allows faculty members to manage notices remotely without requiring direct access to the hardware device. The deployment process also makes it easier to update and maintain the system, ensuring smooth operation and improved accessibility of the digital notice board platform.

6.5.1 Firebase Hosting Configuration

```
{
  "hosting": {
    "public": "public",
    "ignore": [
      "firebase.json",
      "**/.*",
      "**/node_modules/**"
    ]
  }
}
```

This configuration specifies the public folder for hosting and ignores unnecessary files during deployment.

6.6 Code Organization

The implementation code is separated into frontend, cloud, and hardware modules. This modular structure improves readability, makes debugging easier, and supports future system enhancement.

6.7 Conclusion

The implementation of the IoT Based Digital LCD Notice Board successfully combines frontend design, cloud storage, and embedded hardware operation. The home page provides simple navigation, the ESP8266 retrieves notices from Firebase, and the LCD displays them clearly. Thus, the system achieves real-time and paperless communication in an effective manner.

Chapter 7

SYSTEM TESTING

System testing ensures that the IoT Based Digital LCD Notice Board operates according to the specified requirements. The integrated system consisting of the web interface, cloud database, ESP8266/ESP32 microcontroller, and LCD display was tested using the following strategies:

7.1 Testing Strategies

7.1.1 Unit Testing

- **Web Interface Module:** Verified notice entry, update, and delete operations.
- **Cloud Database Module:** Ensured notices are stored and retrieved correctly from Firebase.
- **ESP8266/ESP32 Module:** Verified Wi-Fi connectivity and successful data fetching.
- **LCD Display Module:** Checked correct display of notices and scrolling functionality.

7.1.2 Data Flow Testing

- Verified data flow from Web Interface to Firebase Cloud Database.

- Verified data flow from Firebase to ESP8266/ESP32 via HTTPS/REST.
- Verified communication from ESP to 16×2 LCD using I2C protocol.
- Tested system behavior for valid and invalid network conditions.

7.1.3 Integration Testing

- Notice upload and cloud storage integration tested successfully.
- Cloud update and ESP fetch process verified.
- ESP data retrieval and LCD display integration tested.
- Real-time update reflected without restarting the device.

7.1.4 End-to-End Testing

- Complete process from notice creation to display was tested.
- Verified handling of long messages using scrolling feature.
- Tested system stability during continuous notice updates.

7.1.5 User Interface Testing

- Verified responsiveness of the web interface.
- Verified correct input validation and error handling.
- Ensured secure login access for authorized users.

7.2 Test Cases

Table 7.1: Test Cases of IoT Based Digital LCD Notice Board

Test Case ID	Scenario	Expected Result	Actual Result
1	Faculty adds a new notice	Notice stored in Firebase	Notice stored successfully
2	ESP connects to Wi-Fi	Successful network connection	Connected successfully
3	ESP fetches updated notice	Latest notice retrieved	Notice fetched correctly
4	Display long notice	Message scrolls properly	Scrolling works correctly
5	Invalid login attempt	Access denied	Unauthorized access blocked
6	Multiple updates in short time	System updates in real time	Updates reflected instantly

7.3 Conclusion

Testing confirms that the IoT Based Digital LCD Notice Board functions correctly and meets the defined requirements. The system successfully integrates cloud storage, wireless communication, and embedded display hardware to provide reliable and real-time notice updates.

Chapter 8

RESULTS

The IoT Based Digital LCD Notice Board was successfully designed, implemented, and tested. The system integrates a web-based interface, Firebase cloud database, ESP8266/ESP32 microcontroller, and a 16×2 LCD display. The following screenshots illustrate the working modules of the system.

8.1 Home Page

The Home Page acts as the initial interface of the system. It provides a simple and user-friendly layout that allows faculty members to navigate to the login section. This page introduces the purpose of the digital notice board and ensures smooth access to the system.

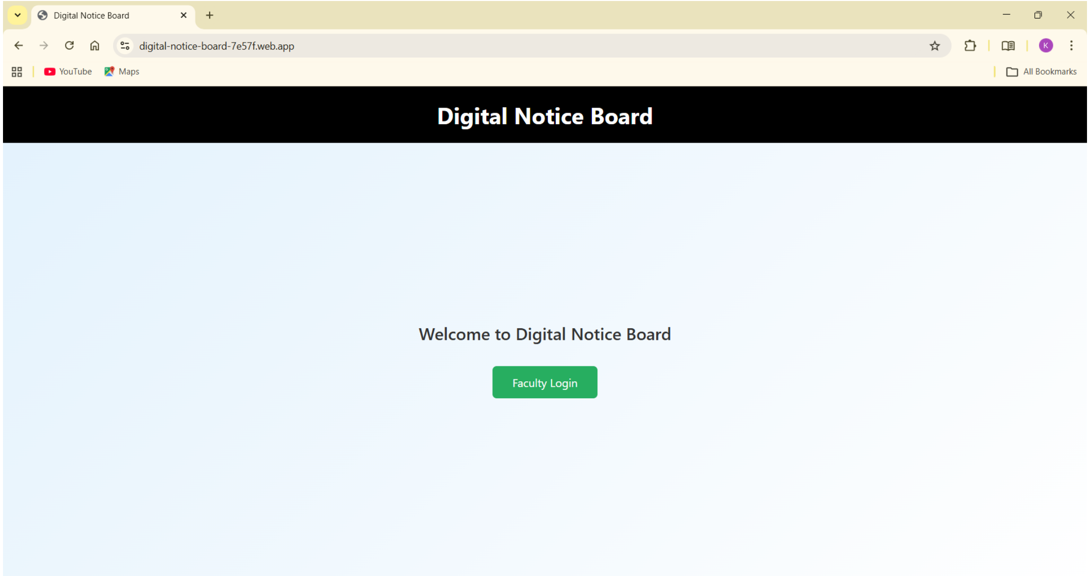

Figure 8.1: *Home Page of IoT Digital LCD Notice Board*

The design ensures clarity and easy navigation. Users can quickly access the login page to manage notices without unnecessary complexity.

8.2 Login Page

The Login Page provides secure authentication for authorized faculty members. Users must enter valid credentials to access the dashboard. This prevents unauthorized users from modifying notices.

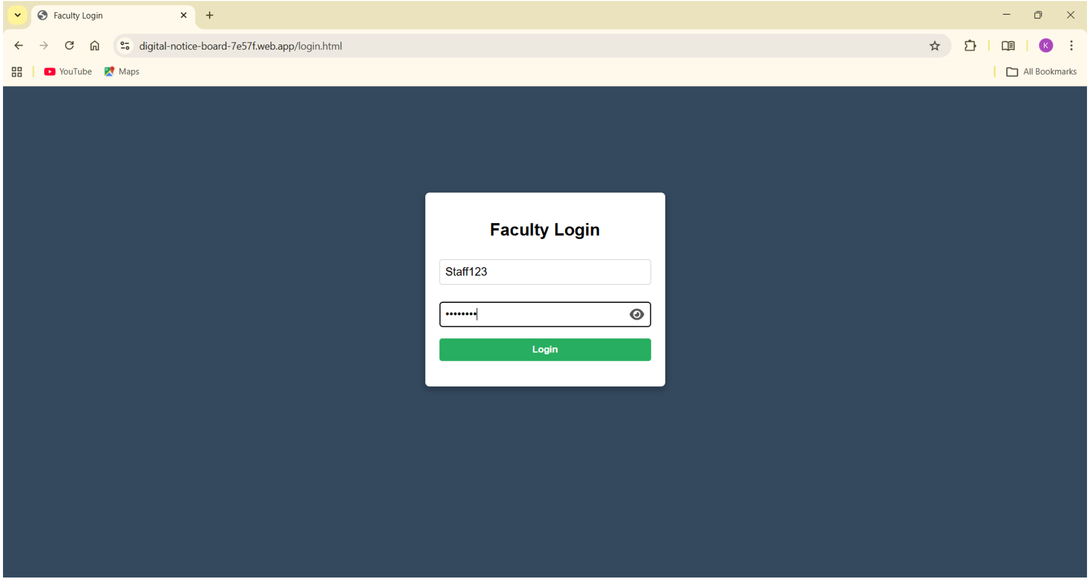

Figure 8.2: *Faculty Login Page*

The system validates the input credentials and grants access only when authentication is successful. If incorrect details are entered, an error message is displayed.

8.3 Faculty Dashboard

The Faculty Dashboard is the main control panel of the system. It allows authorized users to add new notices, update existing notices, or delete outdated information.

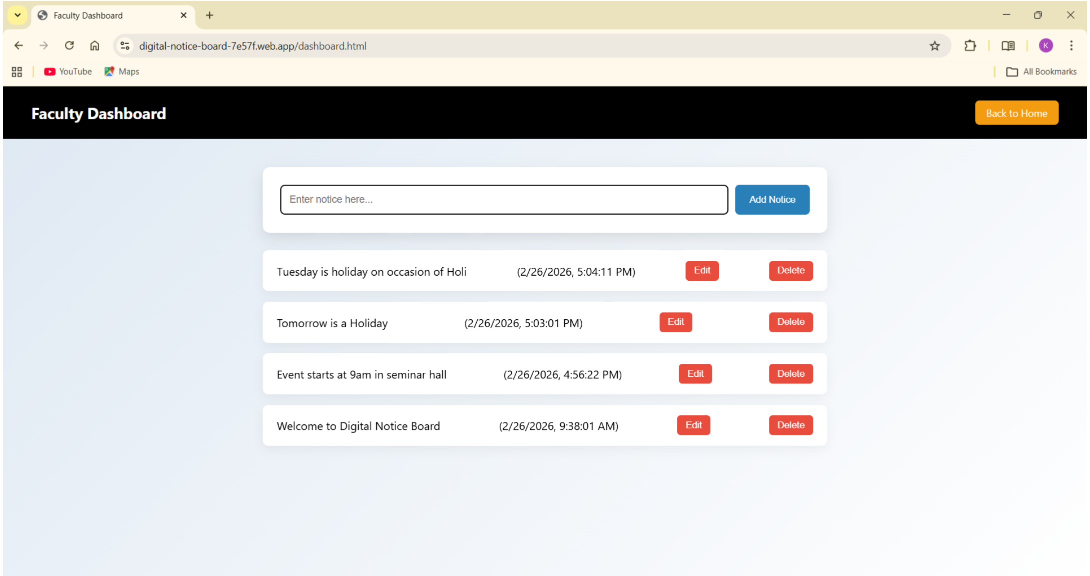

Figure 8.3: *Faculty Dashboard for Notice Management*

Once a notice is submitted, it is stored in the Firebase cloud database. The changes are updated in real time and are immediately available to the display unit. This centralized system eliminates manual notice posting.

8.4 LCD Display Output

The ESP8266/ESP32 microcontroller connects to the Wi-Fi network and continuously monitors the Firebase database for updates. When a new notice is detected, it retrieves the message and sends it to the 16×2 LCD display using I2C communication.

Figure 8.4: *LCD Display Showing Updated Notice*

If the notice length exceeds the LCD width, an automatic scrolling mechanism ensures that the complete message is displayed clearly. This confirms the successful integration of cloud storage, wireless communication, and embedded hardware.

8.5 Conclusion

The results demonstrate that the IoT Based Digital LCD Notice Board operates efficiently. Notices uploaded through the web interface are stored in the cloud and displayed instantly on the LCD screen. The system reduces manual effort, ensures real-time communication, and supports centralized control within institutions.

CONCLUSION

The IoT Based Digital LCD Notice Board was developed to modernize institutional communication by replacing traditional paper-based notice boards with a smart, automated, and internet-enabled system. The project successfully integrates a web interface, cloud database, Wi-Fi-enabled microcontroller, and LCD display to ensure real-time notice updates with minimal manual effort.Throughout the development process, emphasis was placed on system reliability, ease of use, and efficient wireless communication. The web-based interface allows authorized faculty members to manage notices securely, while the ESP8266/ESP32 microcontroller ensures instant retrieval and display of updated information. The implementation of cloud storage enables centralized control and remote accessibility, thereby enhancing overall communication efficiency.This project demonstrates how IoT technology can be effectively applied to create a scalable, cost-efficient, and paperless communication system. The IoT Based Digital LCD Notice Board provides a practical foundation for future improvements such as larger display integration, mobile app connectivity, and advanced notification features. Overall, the system contributes to smarter, faster, and more sustainable institutional communication.

REFERENCES

[1] *"IoT Based Smart Notice Board", Research Inventy International Journal of Engineering and Science*

[2] *"IoT Based Smart Notice Board", ResearchGate Publication*

[3] *"IoT Web Controlled Smart Notice Board using ESP8266", How2Electronics*

[4] *Firebase Official Documentation – Realtime Database*

[5] *ESP8266 Official Datasheet – Espressif Systems*

[6] *Arduino IDE Official Documentation*